ABSTRACT

This project presents the design and implementation of an Electronic Voting Machine (EVM) using the LPC2148 ARM7 microcontroller for real-time embedded voting applications. The proposed system provides a simple, reliable, and low-cost solution for electronic vote recording and result generation. Conventional paper ballot systems involve manual counting procedures that consume significant time, require large manpower, and are prone to human errors and invalid vote entries. To overcome these limitations, the developed EVM automates the voting process using embedded hardware and software techniques. The LPC2148 ARM7 microcontroller acts as the central processing unit and performs vote processing, GPIO control, LED indication, buzzer activation, and UART-based serial communication. Each candidate is assigned a dedicated LED connected through GPIO pins, and whenever a vote is cast, the corresponding LED glows while the buzzer activates to provide immediate visual and audio confirmation. A software delay mechanism is also implemented to prevent repeated voting within a short interval. UART0 communication is configured at 9600 baud rate to transmit final voting results to the UART terminal window in Keil uVision4. The complete system is programmed using Embedded C and simulated in the Keil development environment. Experimental results obtained from GPIO register analysis, Watch Window monitoring, and UART serial output confirm accurate vote counting, correct LED and buzzer operation, and reliable UART communication. The project successfully demonstrates practical ARM7 embedded system concepts including GPIO interfacing, serial communication, real-time event handling, and embedded software implementation. The developed system proves that the LPC2148 microcontroller is suitable for efficient and cost-effective electronic voting applications and embedded systems learning.

KEYWORDS

Electronic Voting Machine, LPC2148, ARM7, UART Communication, Embedded C, GPIO Interfacing, Keil uVision4, Embedded Systems

Problem Statement

Traditional voting systems involve manual vote counting, which is time-consuming, error-prone, and requires significant manpower. There is a need for a low-cost embedded electronic voting system capable of accurate vote counting, real-time vote confirmation, and fast UART-based result generation using the LPC2148 ARM7 microcontroller.

INTRODUCTION

1.1 Overview

Electronic Voting Machines (EVMs) are embedded electronic systems used to record and process votes during elections. Traditional paper ballot systems require manual vote counting, which consumes time and may lead to human errors. Electronic voting systems provide a faster and more reliable alternative.

In this project, an Electronic Voting Machine is implemented using the LPC2148 ARM7 microcontroller. The system records votes for multiple political parties using software-controlled vote selection logic. Vote confirmation is provided through LEDs and buzzer indication. The final voting result is transmitted through UART communication and displayed in the Keil UART terminal window.

The project demonstrates practical implementation of:

- GPIO interfacing
- UART communication
- ARM7 programming
- Embedded C programming
- Real-time embedded control

The complete system is simulated using Keil uVision4 and can also be implemented on LPC2148 hardware.

The proposed system provides:

- Accurate vote counting
- Fast result generation
- Reduced manual effort
- Reliable embedded operation
- Low-cost implementation

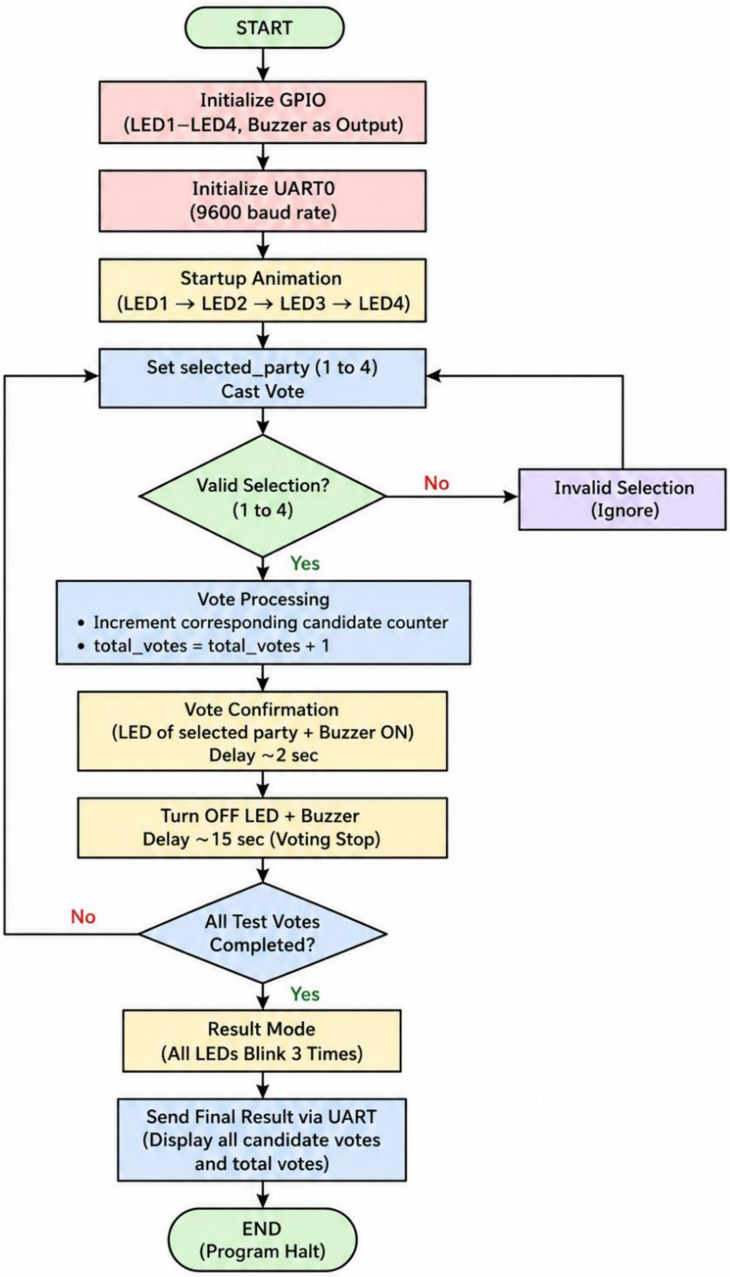
1.2 Need for Electronic Voting Machine

Traditional voting systems suffer from several disadvantages such as:

- Slow vote counting
- Human errors during counting
- Invalid ballot papers
- Large manpower requirement
- Delay in result declaration

Electronic Voting Machines overcome these problems by automating the vote recording and counting process. Embedded systems improve voting accuracy and reduce human intervention.

Figure 1.1: Basic Concept of Electronic Voting Machine



The LPC2148-based EVM offers:

- Fast processing
- Reliable operation
- Simple implementation
- Real-time vote confirmation
- UART-based result display

Thus, Electronic Voting Machines improve election efficiency, reliability, and automation.

1.3 Aim of the Project

The aim of this project is to design and simulate an Electronic Voting Machine using LPC2148 ARM7 microcontroller with LED indication, buzzer confirmation, and UART-based result display.

1.4 Objectives

The major objectives of the project are:

- To interface LEDs with LPC2148
- To implement vote counting mechanism
- To interface buzzer for vote confirmation
- To configure UART0 communication
- To simulate the system in Keil uVision4
- To display final results through UART communication

1.5 Scope of the Project

The proposed system can be used for:

- College elections
- Student voting systems
- Organization voting systems
- Embedded systems laboratory experiments

Future enhancements include:

- Fingerprint authentication
- RFID voting
- GSM communication
- EEPROM storage
- Touchscreen voting

LITERATURE SURVEY

2.1 Existing Voting Systems

Earlier voting systems used paper ballots where voters manually selected candidates. Vote counting was performed manually, requiring large manpower and time.

Traditional voting systems suffered from:

- Human counting errors
- Invalid ballots
- Slow result generation
- High operational cost

Electronic Voting Machines were introduced to improve:

- Speed
- Accuracy
- Reliability
- Election efficiency

Modern EVMs automate vote counting and reduce human intervention.

2.2 Microcontroller-Based Voting Systems

Microcontroller-based voting systems use embedded controllers for:

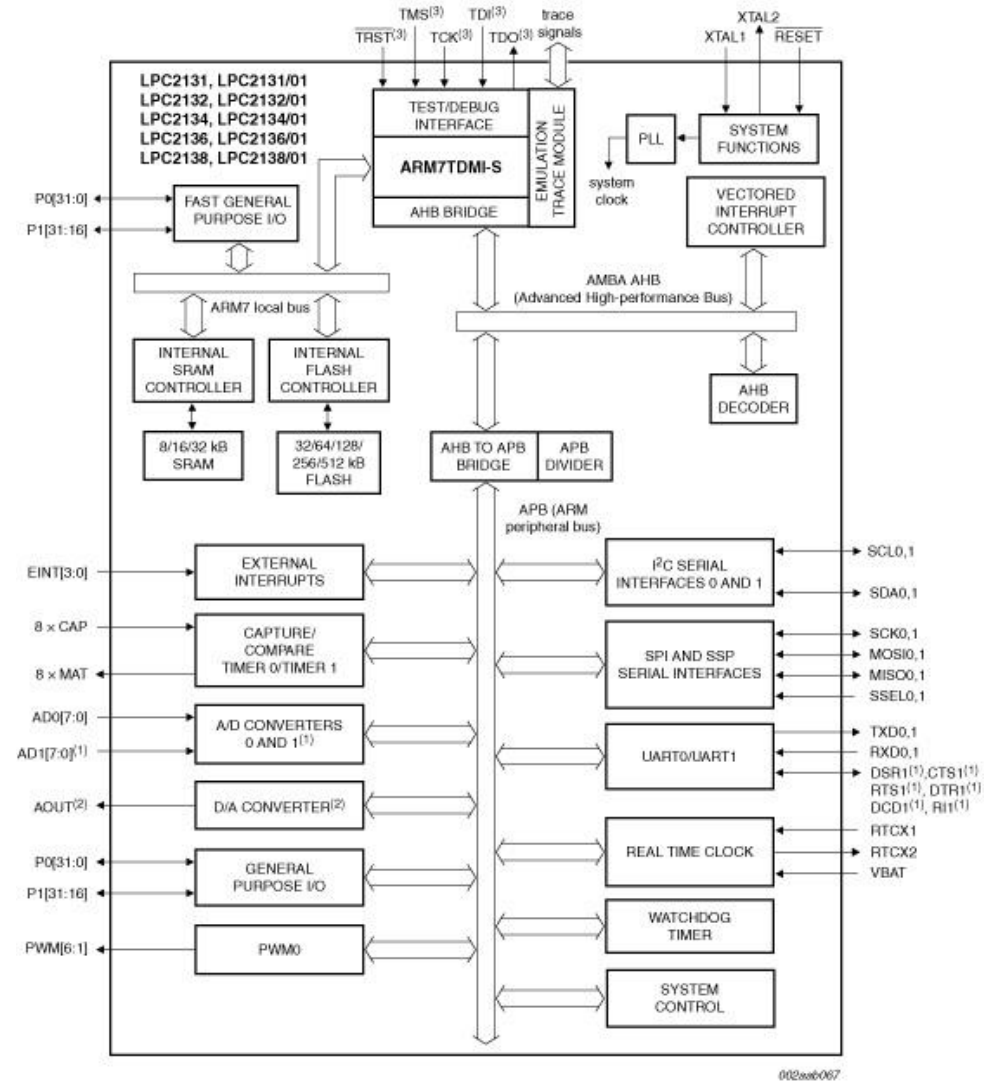
- Vote recording
- Data processing
- Result generation

Among these, ARM7 controllers provide:

- Higher processing speed
- Better memory support
- Multiple peripherals
- Efficient GPIO operation

ARM7 microcontrollers are widely used in embedded applications because of their reliability and real-time performance.

Figure 2.1: LPC2148 ARM7 Microcontroller



2.3 Why LPC2148?

The LPC2148 ARM7 microcontroller is selected because it provides:

- 32-bit ARM7 processing
- Fast GPIO switching
- UART communication support
- Reliable embedded performance
- Large Flash memory

The controller efficiently handles:

- LED interfacing
- Vote processing
- UART transmission
- Buzzer control

Thus, LPC2148 is suitable for Electronic Voting Machine implementation.

HARDWARE DESCRIPTION

3.1 LPC2148 ARM7 Microcontroller

The LPC2148 is a 32-bit ARM7TDMI-S microcontroller developed by NXP Semiconductors for embedded applications. It provides high-speed processing and multiple on-chip peripherals.

Features of LPC2148:

Feature	Specification
Processor	ARM7TDMI-S
Flash Memory	512 KB
SRAM	40 KB
UART	2 UARTs
ADC	14 Channels
DAC	1 Channel
Operating Voltage	3.3V

In this project, LPC2148 performs:

- Vote processing
- LED control
- UART communication
- Buzzer activation

3.2 LEDs

LEDs are used for:

- Party indication

- Vote confirmation
- Startup indication
- Result indication

LED Mapping

LED	GPIO Pin	Party
LED1	P0.16	Candidate_01_PARTY_A
LED2	P0.17	Candidate_02_PARTY_B
LED3	P0.18	Candidate_03_PARTY_C
LED4	P0.19	Candidate_04_PARTY_INDEPENDENT

During vote casting:

- Selected LED turns ON
- LED remains active during confirmation delay
- LED turns OFF automatically

3.3 Buzzer

A buzzer connected to P0.9 provides audio confirmation during vote registration.

Functions of buzzer:

- Vote confirmation
- User acknowledgment
- System indication

Working:

- Buzzer turns ON after vote selection
- Remains active for approximately 2 seconds
- Turns OFF after confirmation delay

3.4 UART Communication

UART0 is used to transmit:

- Candidate vote count
- Total votes
- Final voting result

UART Configuration:

- Baud Rate : 9600 bps
- Data Bits : 8-bit
- Stop Bit : 1
- Parity : None

UART output is displayed in:

- Keil UART #1 Window
- Serial terminal during hardware implementation

3.5 GPIO Pins Used

GPIO Pin	Function
P0.16	LED1 – Candidate_01_PARTY_A
P0.17	LED2 – Candidate_02_PARTY_B
P0.18	LED3 – Candidate_03_PARTY_C
P0.19	LED4 – Candidate_04_PARTY_INDEPENDENT
P0.9	Buzzer
TXD0	UART Transmission

SOFTWARE DESCRIPTION

4.1 Embedded C Programming

Embedded C is used to program the LPC2148 ARM7 microcontroller for implementing vote counting, LED control, buzzer indication, and UART communication.

Major Functions Implemented

Module	Function
GPIO Control	LED and buzzer interfacing
UART Communication	Serial result transmission
Delay Function	Timing generation
Vote Processing	Vote counting logic
Result Display	UART output generation

Registers Used

Register	Purpose
IO0DIR	GPIO direction control
IO0SET	Output activation
IO0CLR	Output reset
U0LCR	UART configuration
U0DLL/U0DLM	Baud rate setup

The modular structure improves readability, debugging, and execution efficiency.

4.2 Keil uVision4 Development Environment

Keil uVision4 is used for:

- Coding
- Compilation
- Simulation
- Debugging

Features of Keil uVision4

Feature	Application
ARM7 Support	LPC2148 programming
Simulator	GPIO and UART testing
Watch Window	Vote count monitoring
UART Window	Serial output display
Debugger	Program analysis

The software enables real-time monitoring of GPIO activity and UART communication.

4.3 Flash Magic

Flash Magic is used to upload the HEX file into LPC2148 hardware.

Flash Magic Operations

Step	Operation
1	Compile program in Keil
2	Generate HEX file
3	Connect LPC2148 hardware
4	Upload HEX file
5	Execute program

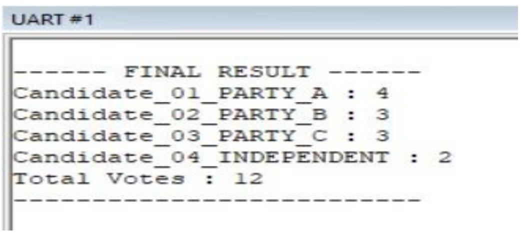
4.4 UART Serial Window

UART0 communication is used to display the final voting result in the Keil UART terminal.

UART Configuration

Parameter	Value
Baud Rate	9600 bps
Data Bits	8-bit
Stop Bit	1
Parity	None

UART Output



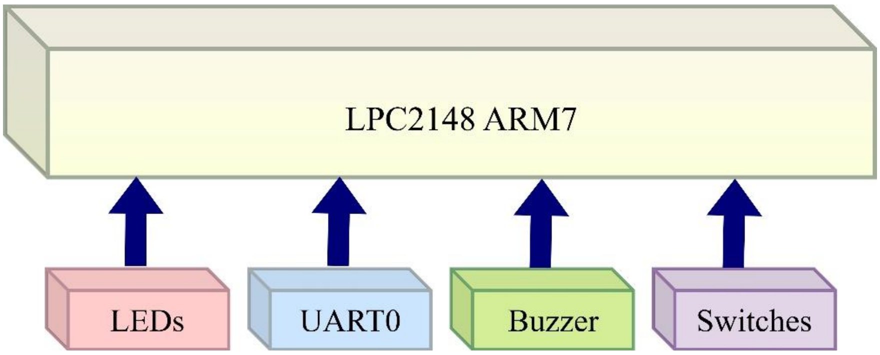
The UART output confirms accurate vote processing and reliable serial communication.

SYSTEM DESIGN

5.1 Block Diagram

The proposed Electronic Voting Machine uses LPC2148 ARM7 microcontroller as the central processing unit interfaced with LEDs, buzzer, UART module, and vote selection mechanism.

System Block Diagram



5.2 Working Principle

The working operation of the system is summarized below:

Step	Operation
1	System initializes GPIO and UART
2	User selects candidate
3	Vote counter increments
4	LED and buzzer activate
5	Delay prevents repeated voting
6	Final result transmitted through UART

The system provides real-time vote processing and reliable embedded control.

5.3 Vote Confirmation Mechanism

After successful vote registration:

- Corresponding LED turns ON
- Buzzer activates
- Delay lock prevents multiple voting

Candidate-wise GPIO Operation

Candidate	GPIO Pin	Indication
Candidate_01_PARTY_A Party	P0.16	LED ON + Buzzer
Candidate_02_PARTY_B	P0.17	LED ON + Buzzer
Candidate_03_PARTY_C	P0.18	LED ON + Buzzer
Candidate_04_PARTY_INDEPENDENT	P0.19	LED ON + Buzzer

The GPIO simulation verified correct LED and buzzer operation during vote processing.

5.4 GPIO Register Analysis

GPIO Registers Used

Register	Function
IO0DIR	Configure output pins
IO0SET	Turn ON LED/Buzzer
IO0CLR	Turn OFF LED/Buzzer

Example GPIO Operations

Turn ON LED and buzzer:
IO0SET = LED1 | BUZZER;
Turn OFF LED and buzzer:
IO0CLR = LED1 | BUZZER;
The GPIO register activity confirms proper embedded hardware interfacing.

PROGRAM IMPLEMENTATION

6.1 Program Modules

The software is divided into different functional modules for efficient operation.

Module	Function
delay()	Timing generation
UART0_Init()	UART initialization
vote()	Vote counting
confirm_vote()	LED and buzzer control
send_result()	UART result transmission

6.2 Delay Function

The delay function is used for:

- LED blinking
- Buzzer timing
- Vote lock delay

```
void delay(int d)
{
   int i,j;

   for(i=0;i<d;i++)
   {
      for(j=0;j<60000;j++);
   }
```

```
}
```

6.3 UART Initialization

UART0 is configured for serial communication.

Parameter	Value
Baud Rate	9600 bps
Data Bits	8-bit
Stop Bit	1
Parity	None

```
void UART0_Init()
{
    PINSEL0 |= 0x00000005;

    U0LCR = 0x83;
    U0DLL = 97;
    U0DLM = 0;
    U0LCR = 0x03;
}
```

6.4 Vote Processing

The vote() function:

- Identifies selected candidate
- Increments vote counter
- Activates LED and buzzer
- Updates total votes

```
switch(selected_party)
{
    case 1:
        Candidate_01_PARTY_A++;
        confirm_vote(LED1);
        break;
}
```

6.5 Vote Confirmation

The confirm_vote() function provides visual and audio confirmation.

```
void confirm_vote(unsigned int led)
{
    IO0SET = led | BUZZER;

    delay(20);

    IO0CLR = led | BUZZER;
```

```
    delay(150);
}
```

6.6 Result Transmission

The send_result() function transmits final voting results through UART communication.

UART Output

Candidate_01_PARTY_A : 4

Candidate_02_PARTY_B : 3

Candidate_03_PARTY_C : 3

Candidate_04_CANDIDATE_04_PARTY_INDEPENDENT : 2

Total Votes : 12

RESULTS AND DISCUSSION

7.1 Experimental Results

The Electronic Voting Machine was successfully simulated in Keil uVision4.

Observations

Operation	Status
Vote counting	Successful
LED indication	Successful
Buzzer activation	Successful
UART output	Successful

GPIO operation:

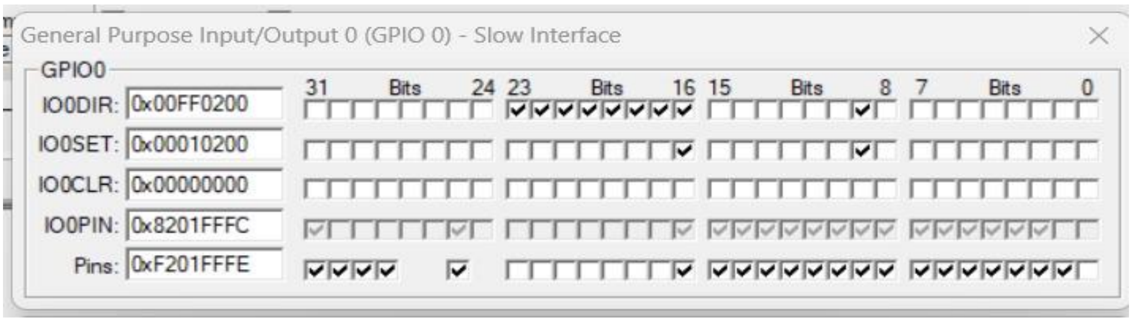
- P0.16 → Candidate_01_PARTY_A Party
- P0.17 → Candidate_02_PARTY_B
- P0.18 → Candidate_03_PARTY_C
- P0.19 → Candidate_04_PARTY_INDEPENDENT

7.2 GPIO Output Verification

Candidate_01_PARTY_A Party Vote

GPIO Pin	IO0SET Value
P0.16	0x00010200

Figure 7.1: GPIO Activity for Candidate_01_PARTY_A Party Vote

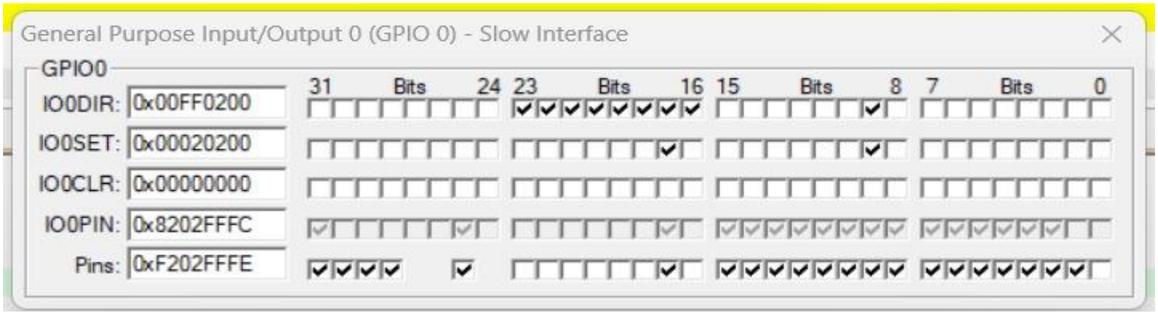


Candidate_02_PARTY_B Vote

GPIO Pin IO0SET Value

P0.17 0x00020200

Figure 7.2: GPIO Activity for Candidate_02_PARTY_B Vote

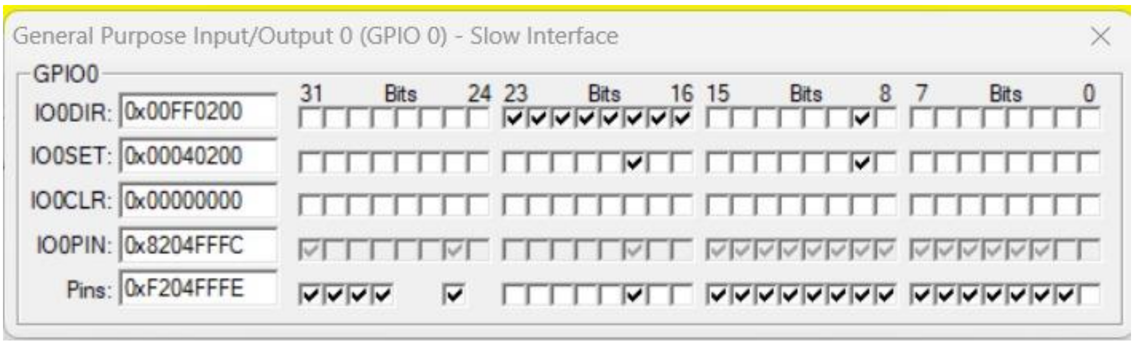


Candidate_03_PARTY_C Vote

GPIO Pin IO0SET Value

P0.18 0x00040200

Figure 7.3: GPIO Activity for Candidate_03_PARTY_C Vote

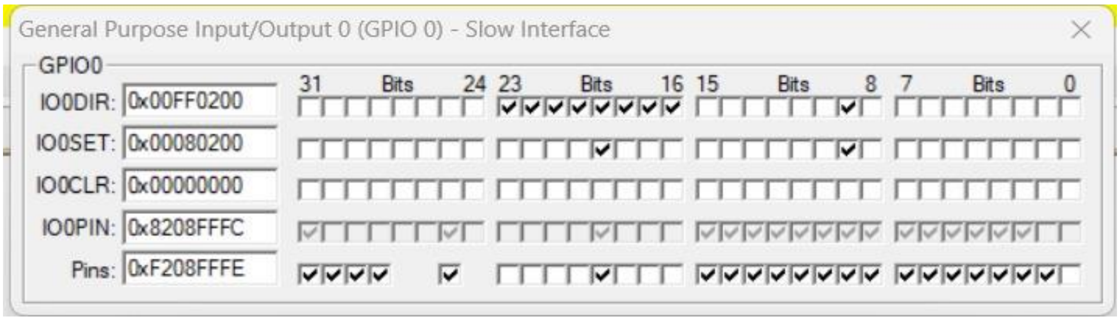


Candidate_04_PARTY_INDEPENDENT Candidate Vote

GPIO Pin IO0SET Value

P0.19 0x00080200

Figure 7.4: GPIO Activity for Candidate_04_PARTY_INDEPENDENT Candidate Vote



7.3 Final Vote Count

Figure 7.5: Final Vote Count in Watch Window

Watch 1

Name	Value	Type
Candidate_01_PARTY_A	0x00000004	int
Candidate_02_PARTY_B	0x00000003	int
Candidate_03_PARTY_C	0x00000003	int
Candidate_04_INDEPENDENT	0x00000002	int
total_votes	0x0000000C	int
selected_party	0x00000001	int
<Enter expression>		

7.4 UART Result Output

Figure 7.6: UART Result Output

UART #1

```
------ FINAL RESULT ------
Candidate_01_PARTY_A : 4
Candidate_02_PARTY_B : 3
Candidate_03_PARTY_C : 3
Candidate_04_INDEPENDENT : 2
Total Votes : 12
--------------------------
```

7.5 Discussion

The LPC2148 ARM7 controller successfully performed:

- GPIO interfacing
- Vote counting
- LED control
- Buzzer indication
- UART communication

The results verified accurate vote processing and reliable embedded system operation.

ADVANTAGES, LIMITATIONS AND FUTURE SCOPE

8.1 Advantages

The proposed LPC2148-based Electronic Voting Machine offers:

- Fast vote counting
- Accurate result generation
- Low-cost implementation
- Simple hardware interfacing
- Reliable UART communication
- Real-time embedded operation
- Easy debugging and simulation

8.2 Limitations

The current system has some limitations:

- No voter authentication
- No permanent vote storage
- No network connectivity
- Limited security features
- Suitable mainly for small-scale voting applications

8.3 Future Scope

The proposed system can be enhanced using advanced technologies such as:

- Fingerprint authentication
- RFID-based voting
- GSM result transmission
- EEPROM/Flash storage
- Touchscreen interface
- Cloud-based monitoring
- Secure encrypted voting

These improvements can make the system more secure, scalable, and suitable for real-world applications.

CONCLUSION

The Electronic Voting Machine using LPC2148 ARM7 microcontroller was successfully designed, implemented, and simulated using Embedded C programming in the Keil uVision4 environment. The system effectively demonstrated real-time embedded operations including vote counting, LED indication, buzzer confirmation, GPIO interfacing, and UART-based serial communication. The implemented software successfully processed votes for multiple candidates and displayed the final election results through the UART terminal window.

The simulation results verified correct operation of LEDs, buzzer, GPIO registers, and UART communication during each voting operation. The project demonstrated the practical implementation of ARM7 embedded concepts such as peripheral interfacing, serial communication, software timing control, and real-time event handling. The overall system performance confirmed that the LPC2148 microcontroller is suitable for low-cost and reliable embedded voting applications.

This project also serves as an effective academic demonstration for learning ARM7 programming, Embedded C development, UART communication, and embedded hardware interfacing. With future enhancements such as biometric authentication, secure data storage, and wireless communication, the proposed system can be extended into a more advanced and secure electronic voting platform.

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